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How to benefit more from business process documentation? Framework for calculation personality - process role fit

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Abstract

Business Process Management (BPM) is perceived as the spanning discipline that largely integrates and completes what previous disciplines have achieved (Brocke and Rosemann, 2014). As such, its integrating activities focused on to discover, model, analyse, measure, improve, optimize, and automate business processes (Weske, 2012). All of these enables implementing performance measurements within the organization, which in effect can lead to decision making based on data rather than people's opinion.

Keywords: BPMN; RIASEC, FFM

1. Introduction

In this paper, we are concentrating on People or rather Process Performers (PP) (Hammer, 2014), one of the BPM key components, according to the classic Six Core Elements of BPM (Brocke and Rosemann, 2014). Taking into consideration personality - process role fit, as a process of the quality measure. The presented approach is also related with the idea of assessing knowledge worker (Davenport, 2014). The approach seeks its business case in reality, especially in an organizational environment where change occurs dynamically, and where is a need to make PP allocation decisions, like in the new ventures or processes redesign. Then, the challenge is faced by managers,

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who need to define tasks, roles, and assign them to the PP. Besides, facing the so-called "halo effect" (Bella et al, 2017), which is the tendency to automatically, positively or negatively, assign personality traits based on the first impression. It is also an opportunity, to utilize psychology frameworks and decades of findings in fields of business performance and its relation to personality. That, in effect, allows capturing in numbers the notion, that no one could be predisposed to all variety of tasks. Moreover, having benefits in one type of task there is a chance to having discomfort in another (Niknafs et al., 2013).

2. The origin of the problem

The concept of personality - process framework started its existence as a result of a project implemented in a commercial bank with the HR department. The organization already had some experience and knowledge in supporting decision making based on personality in the areas: team-building process and selection of candidates for positions. Although in a very simplified manner in terms of numbers, it brought satisfactory results. Another important aspect is that the Organization places special emphasis on Process Management and observes high dynamics in selected areas. Either there are frequent changes of roles, work divisions in existing ones or new ones are created with a new staff in a rapid manner. Hence, the business solution, which consists of developing a method that allows apriori to identify potential risks in personnel decisions, is needed.

The first stage, which coincides with the scope of this article, is to develop the concept of a mathematical model capturing the relationship of the personality profile to the efficiency and performance of the task. The purpose of the stage came from the fact that before we proceed to process more complex things, such as team building, modeling should first be subject to the basic relationship Profile - Task. The product of the model from the user's perspective is the classification of the employee profile into one of the classes determining the degree of matching to the tasks or their group and indication of the optimal decision to choose from the profile group. Ultimately, we are heading to a method that allows two things (1) to optimize the business process in terms of the nature of the work performed: new work distribution, (2) give a given profile base, an indication of the optimal profile - role in the business process: new team creation or a new division of roles.

2.1. Why Processes' and BPMN Perspective?

Although, in our opinion key enabler of this method is a business process (BP) expressed in Business Process Management Notation (BPMN). The Business Process Modeling Notation (BPMN) has recently emerged as the industry-standard notation for modeling business processes (Dumas et al., 2018). BPMN has over the last years has been met with the diversity of needs; i.e., to create models to be understandable both for humans and machines, for sense-making, quality management, simulation, and activation, although an approach like BPMN is not necessarily suitable for all usage areas e.g. lack of support for routine work (Kossak et al., 2014). In terms of BP assessment the BPMN has following benefits: (i) allows to use build-in and additional parameters to describe tasks and its personality requirements, (ii) all BP can be visualized and stored in XML format and simulated according to implemented BPMN engine (Geiger et al., 2018), (iii) highly efficient integrator between various type of needs and stakeholders. Overall, Business Process Managers and Business Analysts could enhance their toolset by using this method in analyzing and modeling (BP). Value proposition could be described as (i) verify set and combination of profiles vs BP roles and its tasks to choose an optimal combination, (ii) verify role and its tasks, as it could have, from personality perspective conflict requirements, (iii) verify possible hypothesis by process redesign.

2.2. Goal and Organization

This paper explains initial progress in the framework for calculation personality-process role fit development. The next section covers psychology theory and background with a special focus on personality as an effective component and personality models in the business. A compact version of the framework as an approach for calculation personality work fit is proposed as the basis of the discussed framework. The proposed framework is aimed to provided answers for question: What PP's personality is the best fit for each stage of the business process? Based on process documentation of the enterprise model, it seems possible that the framework, regardless of the type of business process and the language of the, could be useful. In the article, we hypothesize that we can effectively solve personality - process role fit problem based on Big Five and Holland personality models.

3. Theoretical background: Personality, Environment and Job Performance

Personality is defined as the character sets of behaviors, cognitions, and emotional patterns that evolve from biological and environmental factors (Sadock et al, 2017). As a work performance impact factor, it is been subject to investigation from many years (Le et al., 2011; Tisu et al, 2020). The main idea behind all this theory states that an employee's satisfaction with a job, as well as effectiveness, depending on the degree to which the individual's personality matches his or her occupational environment. Researchers approached it from many directions, although in terms of effectiveness seeking, one area of studies is highly interesting. It argues that personality could be a valuable predictor of job performance, especially by considering various job positions and their individual specifications. By scrutinizing top performers, a certain correlation between its traits and job characteristics have been found. We believe, that this opens new possibilities in assessing BP and deliver additional quality measures. Key benefits origin from perceiving and emphasizing BP tasks, as a separate activity with individual personality preferences, that assessment of it versus PP profile is possible.

In our research, we follow the main idea of personality psychology that personality can be described by a set of traits, i.e. fixed set of patterns in how a person behaves, feels, and thinks (Matthews et al., 2009). Based on these traits we can summarize, explain, and predict how a person will act in different situations. Even though different trait-based theories had been proposed it was not until the Five-Factor Model (FFM) gained more widespread acceptance and interest.

3.1. Five Factor Model

In terms of personality models, one clearly dominates - The Big Five personality traits, also known as the five-factor model (FFM) or OCEAN model. The FFM is a widely known methodological tool (and taxonomy) in counseling psychology research for grouping personality traits (Baldwin and Del Re, 2016). The FFM suggests five personality traits (Storme et al., 2016): Extraversion (to be sociable, active), Agreeableness (to be softhearted, trusting), Conscientiousness (to be organized, reliable), Emotional Stability (to be calm, relaxed), and Openness (to be curious, creative). The FFM is a hierarchical organization of personality traits. Under each of the basic five factors, there are six sub-facets for a total of 30 different personality facets.

FFM has a long history but what is interesting, thanks to the digital age and big data, it could be utilized and study from many perspectives. Signs of FFM model application we can find in social-media advertising, games, interactive storytelling, and more (Matthews et al, 2009). All of the findings put in the light personality perspective as a very promising and informative to utilize. Moreover, considering that it is feasible to obtain personality profiles, either from a dataset from science open databases or from one of the overall available online forms, we found FFM as an optimal framework to describe PP personality profiles.

3.2. Holland Model

However, it seems highly challenging to define the personality requirements of BP's tasks in the FFM framework. Mainly, as it is not so intuitive to perceive activity or task from the FFM perspective and second, it is hard to tackle automated processing of the BPMN within FFM as there are no promising rules to apply. It leads to a need for the adoption of another model, which can be helpful in a more convenient way of a task description. As for analyst and for automated processing as well.

The Holland vocational model assumes that people develop habitual or preferred methods of dealing with environmental tasks (Holland, 1997). As researches point out, by helping to generate core knowledge related to career development, assessment, and practice, Holland's theory and research have contributed in innumerable and significant ways to the field of counseling psychology (Nauta, 2010). Holland's RIASEC model of occupational environments defined following occupational codes: Realistic (Doer), Investigative (Thinker), Artistic (Creator), Social (Helper), Enterprising (Persuader), Conventional (Organizer). It is important to emphasize that, any work environments can be classified by their affinity to a combination of the RIASEC types, and Holland codes are often used to describe them as well as a personality. Many different measures of the RIASEC model have been developed over the decades. The common elements of all Holland measures are that they present participants with a list of specific activities, asking whether they would like to do them versus would dislike or be indifferent to doing them (examples of “enterprising” activities include “promote a product” and “start my own business”) (Sheldon et al., 2020). Each of the six vocational types reflects a distinct constellation of interests, skills, and dispositions, although the six types can coexist and overlap with each other. Moreover, research has shown meaningful convergence between the six types and FFM traits (Tokar & Swanson, 1995).

John Holland’s theory of career orientations advises people to select careers that are congruent with their personalities (Holland, 1997). We go further and recommend them select tasks that are congruent with their personalities. At the same time, we give the users the ability of intuitive tasks personality requirements description. In our opinion, a better option is to define the task as Investigate, which is affected by three FFM traits, rather than using it directly and state which one to which degree and with what direction.

The research area focused on personality dimensions and job performance is dynamic. The classic work (Barrick and Mount, 1991) provided interesting results, however, the job performance is described by the task performance and creativity which are not standardized. In our opinion, only work connecting the FFM with Holland Occupational Codes can be considered valuable. According to (Holland, 1997) the people’s vocational capabilities flow from their life history and personality.

Depending on the methodology, we can get different results to determine the relationship between the FFM’s and RIASEC. The research started from (Larson et al., 2002; Barrick et al., 2003; Mount et al., 2005) with univariate meta-analyses of the relationship between RIASEC interests and FFM personality dimensions. The last results come from (Hurtado Rúa et al., 2019) with multivariate meta-regression meta-analysis.

At the current stage of research, there are many open issues. One of them is a small value of relationships (most are around .25) for both the correlation between the character trait (FFM) and the performance, as well as between FFM and Holland. However, we base the validity of our model on the fact that there are N non-zero correlations, which can be numerically recognized and thus processed. We can also add that we are striving for a model that will ultimately support non-linearity. To sum up, we can confirm that we cannot find the research that relates FFM and Holland models with business process documentation.

4. Literature overview

The relationship between personality and job performance has been a frequently studied research topic over the past century and to embark on new directions for research. Especially the emphasis is put on the relationship between the FFM personality traits and job performance (Barrick& Mount, 1991). The current evidentiary base on the use of personality measures for understanding individual behavior in the workplace has led to both enthusiastic support (Hurtado Rúa et al., 2019) and strong criticism (Morgeson et al., 2007). In the next paragraphs, we will analyze the different stages of understanding personality-job performance relationships:

Generalization: FFM traits affected overall job performance. To be more precise, many researchers support the findings that conscientiousness is a valid predictor across performance measures in all occupations studied (Barrick et al., 2001). Emotional stability was also found to be a generalizable predictor when overall work performance was the criterion, but its relationship to specific performance criteria and occupations was less consistent than was conscientiousness (Barrick et al., 2001).

Specialization: FFM traits affected job performance depending on job context and environment. The personality factor of extraversion has been associated with performance in some occupations (e.g., sales), and it has been one of the most consistent personality predictors of leadership (Blickle et al., 2015). Moreover, higher levels of extraversion are linked to performing more efficiently when working according to a schedule or project plan, and higher levels of openness are linked to preferring to take responsibility for a whole project and not individual parts (Feldt et al., 2010). In addition, employers engaged in a project-oriented job with more intense personalities (scoring higher on the factors of extraversion and openness) were found to be more likely to be younger, to prefer doing multiple things at the same time, and to prefer contributing to some part of a project instead of working with it from the start to the finish (Feldt et al., 2010).

Bidirectionality: same trait positive and negative correlations on job performance. Generally, (Tett et al. 1999) point out the possibility of bidirectionality of the link between trait and trait-expressive behaviors (Extraversion, for example, may be required in some jobs and introversion in others). In other words, the presence of a trait-relevant situation predicts a positive relationship between personality traits and trait-expressive behaviors, whereas a lack of a trait-relevant situation may reverse this relationship (Li et al., 2018). Additionally, (Helle at al., 2018) study indicates that the FFM personality domains and facets can provide information regarding various types of workplace behaviors and these relationships are not always reciprocal. For example, if antagonism is associated with counterproductive behaviors, it does not necessarily indicate that agreeableness will be associated with positive or citizenship oriented behaviors. Conscientiousness, on the other hand, is associated with counterproductive behaviors (negatively) and organizational behaviors (positively).
Nonlinearity: FFM traits affected job performance in a nonlinear fashion. The last iteration of the research hypothesis assumes that the personality and job performance relationship is non-linear and asymptotic (Blickle et al., 2015). In particular, some researchers have suggested that the descriptors of normal range personality may be related in a curvilinear fashion to occupational performance (Walmsley et al., 2018). The presence of robust nonlinear personality–performance relationships would motivate rethinking about the methods used to estimate the utility of personality measures in workplace settings.

5. Framework

The presented framework is developed to have the capability of performing the analysis of optimal choice in terms of personality profile - BP role assignment (Figure. 1). For the given set of personality profiles P, tasks T, and correlation matrix, between FFM (OCEAN) and Holland (RIASEC), it calculates fit function, which expresses the degree of which the profile is meeting the requirements. Besides calculations, fuzzy logic transformations enable presenting the final results, by showing the profiles' class, which is a more convenient way of communication from the users' perspective.

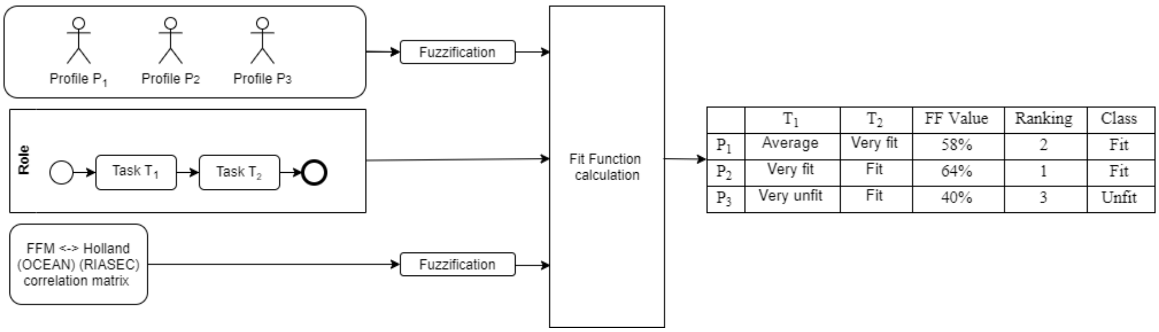

Figure 1. Method overview

5.1. Profiles, Tasks and Roles

The personality indicators are described by the FFM and expressed in the form of a vector with five elements, and each of them directly corresponds to the FFM OCEAN's parameters. For person i the FFM's vector Pi can be described Pi= [Oi, Ci, Ei, Ai, Ni] where Oi, Ci, Ei, Ai, Ni $\in$ <0,100>. Ultimately processed values of personality traits are the first subject to fuzzification, which allows among others, addressing the phenomenon of non-linearity. For example, we introduce an additional class for 85 - 100 ranges, where the matching value resulting from the extreme values of the feature is reduced.

From another initial step, we obtained for each Task j vector Tj = [Rj, Ij, Aj, Sj, Ej, Cj] where Rj, Ij, Aj, Sj, Ej, Cj $\in$ <0,1> and elements Rj, Ij, Aj, Sj, Ej, Cj are corresponding to appropriate position in the Holland Occupational model. The values of individual elements add up to 1, this approach reflects the view that each task can have different components and accents. The set of tasks is included in Roles, R = [T1, T2, ..., Tn] and after averaging the values for individual tasks, it can express the overall personality requirement.

5.2. Correlations between Holland Occupational Codes and Five Factor Model (OCEAN-RIASEC)

An important component of the calculation is the impact of personality traits on the nature of the task being performed. Moreover, it is necessary to address two issues arising from literature and research. First, its bi-directional specificity, in the sense that in one case it can have a positive and in another negative impact on work efficiency. Secondly, the varying degree of influence that personality traits have on each other for a particular type of task.

		Holland Occupational Codes (RIASEC)					
		Realistic	**Investigative**	**Artistic**	**Social**	**Enterprising**	**Conventional**
Five Factors Model (FFM)	**Neurotic**	low-	high-			low-	
	Extroversion		low-		low+	high+	
	Openness		low+	high+	high+		
	Agreeableness			high+	low+	low-	
	Conscientiousness			low-	high+	low+	low+

Table 1. The relationship matrix between personality described in Five Factors Model and tasks expressed in Holland Occupational Codes

The starting point for us is the results mentioned in the review of the meta-analysis literature, indicating different degrees of correlation between the personality trait and task specificity. Based on this, we assumed the assignment to one of the classes based on the value of the correlation coefficient p.

The results are summed up in Table 1 with a correlation matrix (M) between FFM (OCEAN) and Holland (RIASEC). For example (based on Table 1) the BP with a high level of extroversion will be punished in an Investigative kind of task, while he would be awarded in Enterprising task.

5.3. Fit function

The output fit function (FF) gives the value of the degree of matching the profile P to the role R, which is the average value of matching for all related tasks. In turn, matching to a particular task is the average value of matching personality traits to individual components of the task from the Holland model (RIASEC). For each component of the task, the correlation matrix shows which characteristics, to what extent and sign, have an impact on efficiency.

5.4. Use case

One the of project's cases was analyzing BPMN with a specific role and related job-position description of BPMan (Figure 2). Based on an analysis of the documentation (BPMN, task description) an expert defines the values of RIASEC for every task (see Figure 3). As this documentation is part of the company know-how its publicity is limited by a non-disclosure agreement. Alternatively, we used as an example (BPMan tasks and its description) from Job' Hero Catalogue.

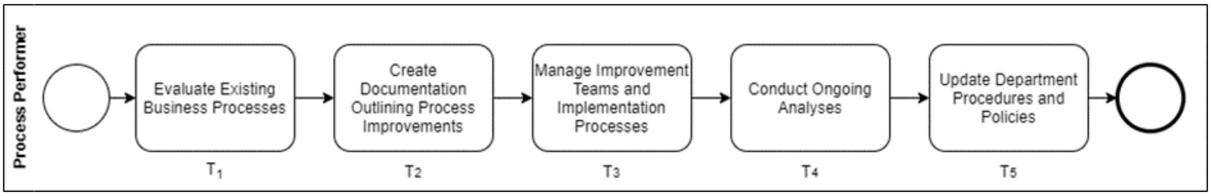

Figure 2. Example BPMN process for BPMan's tasks

	R	I	A	S	E	C
T_1	0	0,8	0	0	0	0,2
T_2	0	0	0,3	0	0	0,7
T_3	0	0	0	0,4	0,6	0
T_4	0,1	0,8	0	0	0	0,1
T_5	0,2	0	0	0,3	0,3	0,2

	O	C	E	A	N
P_1	80	70	40	40	52
P_2	35	71	90	69	54

	T_1	T_2	T_3	T_4	T_5	FF Value	Class	Ranking
P_1	Very Fit	Very Fit	Fit	Very Fit	Fit	67,7%	Very Fit	1
P_2	Unfit	Very Fit	Fit	Unfit	Average Fit	54,6%	Average Fit	2

Figure 3. Expert RIASEC 's assessment of the example BPMN model from Fig. 2

For example for task T1 (Figure 2): Evaluate Existing Business Processes the expert identified the following actions: to complete this main duty, Process Managers break down various business processes with flowcharts, manuals and other documentation outlining current practices. They get the big picture by assembling this data and studying ways to make improvements to one or many steps to increase productivity, reduce costs, improve time management or make needed changes to other aspects of the process. According to the relationship matrix (Table 1) and expert assessment (Figure 3), the candidate profile fit to the task T1 will be affected by the value of four different candidate's FFM traits. The productivity in the area of the Investigation component of T1 is affected by a candidate's Neuroticism, Extraversion and Openness, while Conventional is impacted by a candidate's Conscientiousness. We test the framework on the real personality Profiles. P1 Business Analyst, P2 HR Business Partner. The results show clear differences in matching between profiles.

6. Conclusion

Despite many decades in the field of personality research, it seems that there is still a lack of universal explicitness in many aspects. Particularly important in terms of the problem from the side of creating a mathematical model. On the other hand, one cannot deny the existence of certain significant correlations, the use of which, while maintaining certain restrictions, can be successfully processed and the conclusions valuable. Finally, we recognize that in the context of assessing the method, it has potential but also limitations, so the following conditions should be taken into account: (1) Personality in itself is not the ultimate predictor of behaviour in a particular situation. It is also dictated by the context. One can, therefore, speak only of some weaker or stronger tendencies than determinism. (2) Personality is not the only criterion affecting the effectiveness of specific work. It needs to be mention things like cognitive skills (IQ), knowledge, or experience. (3) The mere representation of a personality in numerical form based on a self-reporting questionnaire is subject to a structural error and term separately discussed in the psychology assessment field. (4) There are strong reasons to recognize the impact of certain personality traits on the overall work performed. Especially Conscientiousness and Neuroticism show a general impact on most of the work performed. (5) Moreover, in certain cases, bidirectional applications are worth noting. In the sense that the trait can have a potentially positive or negative impact depending on the nature of the work. Due to the change in the sign of the impact of a particular personality trait, differences between profiles may be significant. (6) Due to the distribution of data, wherein the vast majority of values the personality traits are in the medium range, the differences in matching are small. The method works more effectively outside this range, effectively capturing extreme cases. (7) The use of the Holland model turned out to be extremely effective. For those using the method, the description of tasks, provided for the classification of work taxonomy, is accessible and intuitive. Especially compared to the need to define requirements through the FFM model, where it is difficult to express the context of the task.

6.1. Future work

After determining the mathematical model, we intend to go in two directions: (1) Approach to classic assignment problem defined by the question: how to assign n items (e.g. jobs) to m machines (or workers) in the best possible way. In our case, it will Profiles - Roles in BP. The assignment problem consists of finding a maximum weight matching (or minimum weight perfect matching) in a weighted bipartite graph. (2) Work distribution among BP Roles taking into consideration personality requirements. We mean process redesign, i.e. tasks similar to one role and avoiding contradictory ones. We optimize the distribution of tasks for the given personalities. At the same time, it is important to emphasize the need to track research on the non-linearity of the trait - work efficiency relationship. Moreover, each basic FFM factors is divided into six sub-facets for a total of 30 different personality facets. The new level of detail is of key importance for the quality of the framework.

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